



**College of Computing and Informatics  
Department of Computer Science**

**Course Syllabus**

# **Course Title**

# **Foundations of Data Science**

**Course Code**  
**1501564**

**Spring 2024/2025**

## Course Syllabus

|                                 |                             |                        |         |                  |               |
|---------------------------------|-----------------------------|------------------------|---------|------------------|---------------|
| Course Title                    | Foundations of Data Science |                        |         |                  |               |
| Course Code                     | 1501564                     |                        |         |                  |               |
| Credit Hours                    | 3                           | Contact Hours          | 3       |                  |               |
| Course Category                 | Program required            |                        |         |                  |               |
| Course Type                     | Compulsory                  |                        |         |                  |               |
| Pre-requisite(s)                |                             |                        |         |                  |               |
| Co-requisite(s)                 |                             |                        |         |                  |               |
| Semester                        | Spring 2024/2025            |                        |         |                  |               |
| Instructor Name                 | Email                       | Classroom              | Section | Session Type     | Time          |
| Isam Mashhour Hasan Al Jawarneh | ijawarneh@sharjah.ac.ae     | M4: Business (Men)-110 | 61      | Lecture Graduate | W 1700 - 1945 |

### Course Description:

Data science is an interdisciplinary field that provides tools to extract insights from data in various forms, either structured or unstructured. Data science course provides the theories, strategies, and tools to understand and apply the following topics: data preparation, data cleaning & integration, data analysis, classification, clustering, text analysis, and visualization.

### Course Objectives/Goals:

The goals of this Course are to:

- Define the fundamental concepts of data science fundamentals.
- Master a data science tool and apply it to discover knowledge from data.
- Define the advanced concepts of data science.
- Design, develop, and present a data science algorithm.

### Course Learning Outcomes:

By the end of successful completion of this Course, the student will be able to:

1. Apply data preparation techniques to clean and integrate the data.
2. Write and apply scripts to analyze and visualize data.
3. Apply statistical methods to estimate unknown parameters and assess data science model.
4. Apply appropriate classification & clustering techniques to a problem
5. Analyze, design, and implement data science algorithms to create models for solving real-world problems.

### Alignment of Course Student Learning Outcomes to Program Student Learning Outcomes

| Program Learning Outcomes (PLOs)                                  | Course Learning Outcomes (CLOs)                                         |
|-------------------------------------------------------------------|-------------------------------------------------------------------------|
| Apply computing models to devise solutions to data science tasks. | - Apply appropriate classification & clustering techniques to a problem |
| Apply computing models to devise solutions to data science tasks. | - Apply statistical methods to estimate unknown parameters and          |

| Program Learning Outcomes (PLOs)                                                                                           | Course Learning Outcomes (CLOs)                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                            | assess data science model.                                                                                                                                                                                                                                                                  |
| Apply computing models to devise solutions to data science tasks.                                                          | <ul style="list-style-type: none"> <li>- Apply data preparation techniques to clean and integrate the data.</li> <li>- Write and apply scripts to analyze and visualize data.</li> <li>- Apply statistical methods to estimate unknown parameters and assess data science model.</li> </ul> |
| Transform information to discover relationships and insights into complex data sets.                                       | <ul style="list-style-type: none"> <li>- Write and apply scripts to analyze and visualize data.</li> </ul>                                                                                                                                                                                  |
| Create models using proper techniques and methodologies of abstraction that can be automated to solve real-world problems. | <ul style="list-style-type: none"> <li>- Analyze, design, and implement data science algorithms to create models for solving real-world problems.</li> </ul>                                                                                                                                |

### Weekly Distribution of Course Topics/Contents

| Week | Topic                                                                                                                                         | Comments* | CLOs |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------|------|
| 1    | Course overview & Introduction to big data                                                                                                    |           | 1    |
| 2    | <ul style="list-style-type: none"> <li>- Data Science</li> <li>- Data Science Definition</li> <li>- Datafication</li> </ul>                   |           | 1,2  |
| 3    | Python for Data Science: Intro - Tables - Data Types - Building Tables                                                                        |           | 1,2  |
| 4    | Python for Data Science <ul style="list-style-type: none"> <li>- Distributions/Charts</li> <li>- Histograms</li> <li>- Functions</li> </ul>   |           | 1,2  |
| 5    | Python for Data Science <ul style="list-style-type: none"> <li>- Conditionals and Iteration</li> <li>- Groups, Pivots, Joins</li> </ul>       |           | 1,2  |
| 6    | Exploratory Data Analysis (EDA) <ul style="list-style-type: none"> <li>- Sampling</li> </ul>                                                  |           | 1,2  |
| 7    | Testing Hypotheses <ul style="list-style-type: none"> <li>- Comparing Distributions, Decisions/Uncertainty</li> </ul>                         |           | 1,2  |
| 8    | Midterm Review                                                                                                                                |           | 3    |
| 9    | Estimation <ul style="list-style-type: none"> <li>- Confidence Interval</li> <li>- Interpreting Confidence</li> </ul>                         |           | 3    |
| 10   | Use of a Mean <ul style="list-style-type: none"> <li>- Center and Spread, The Normal Distribution</li> <li>- Sample Means</li> </ul>          |           | 3    |
| 11   | Prediction and regression <ul style="list-style-type: none"> <li>- Designing Experiments, Correlation</li> <li>- Linear Regression</li> </ul> |           | 5    |
| 12   | Prediction and regression <ul style="list-style-type: none"> <li>- Least Squares</li> <li>- Residuals</li> </ul>                              |           | 4,5  |
| 13   | Classification <ul style="list-style-type: none"> <li>- k-Nearest Neighbor</li> </ul>                                                         |           | 4,5  |

| Week | Topic                                              | Comments* | CLOs    |
|------|----------------------------------------------------|-----------|---------|
| 14   | Research paper submission and Project presentation |           | 5       |
| 15   | Research paper submission and Project presentation |           | 5       |
| 16   | Final exam                                         |           | 1,2,3,4 |

### Information on out-of-class assignments with due dates for submission

| Assignment/Activity | Due Date                       | Comments |
|---------------------|--------------------------------|----------|
| class activity      | weeks 4,10,14                  |          |
| midterm exam        | As per the university calendar |          |
| Project             | Week 15                        |          |
| Final Exam          | As per the university calendar |          |

### Students' Assessment:

Students are assessed as follows:

| Assessment Tool(s)** | Date | Weight (%) |
|----------------------|------|------------|
| Midterm Exam         |      | 20         |
| Projects             |      | 35         |
| Class Activity       |      | 15         |
| Final Exam           |      | 30         |
| <b>Total</b>         |      | <b>100</b> |

### Course Outcome Assessment Plan:

| Course LOs                                                                              | Teaching/Learning Method(s)                    | Assessment Tool(s)          | Performance Indicators | UoS competencies                                           | Other competencies |
|-----------------------------------------------------------------------------------------|------------------------------------------------|-----------------------------|------------------------|------------------------------------------------------------|--------------------|
| Apply data preparation techniques to clean and integrate the data.                      | - Lectures<br>- Exercises                      | - Exams<br>- Class Activity | Marks >= 80%           | - Critical and analytical thinking<br>- Problem solving    |                    |
| Write and apply scripts to analyze and visualize data.                                  | - Lectures<br>- Exercises                      | - Exams<br>- Class Activity | Marks >= 80%           | - Critical and analytical thinking<br>- Problem solving    |                    |
| Apply statistical methods to estimate unknown parameters and assess data science model. | - Lectures<br>- Exercises<br>- Research papers | - Exams<br>- Class Activity | Marks >= 80%           | - Critical and analytical thinking<br>- Oral communication |                    |

| Course LOs                                                                                               | Teaching/Learning Method(s)                                                                                  | Assessment Tool(s)                                                                  | Performance Indicators | UoS competencies                                                                                                                                                                                                                                                                           | Other competencies |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
|                                                                                                          |                                                                                                              |                                                                                     |                        | <ul style="list-style-type: none"> <li>- Written communication</li> <li>- Teamwork or collaboration</li> <li>- Innovation / creative thinking</li> <li>- Problem solving</li> </ul>                                                                                                        |                    |
| Apply appropriate classification & clustering techniques to a problem                                    | <ul style="list-style-type: none"> <li>- Lectures</li> <li>- Exercises</li> <li>- Research papers</li> </ul> | <ul style="list-style-type: none"> <li>- Exams</li> <li>- Class Activity</li> </ul> | Marks >= 80%           | <ul style="list-style-type: none"> <li>- Critical and analytical thinking</li> <li>- Oral communication</li> <li>- Written communication</li> <li>- Teamwork or collaboration</li> <li>- Problem solving</li> </ul>                                                                        |                    |
| Analyze, design, and implement data science algorithms to create models for solving real-world problems. | <ul style="list-style-type: none"> <li>- Research papers</li> </ul>                                          | <ul style="list-style-type: none"> <li>- Projects</li> </ul>                        | Marks >= 80%           | <ul style="list-style-type: none"> <li>- Critical and analytical thinking</li> <li>- Oral communication</li> <li>- Written communication</li> <li>- Teamwork or collaboration</li> <li>- Leadership skills</li> <li>- Innovation / creative thinking</li> <li>- Problem solving</li> </ul> |                    |

### Scope and Nature of Assessment Tools:

#### Scope and Nature of Assessment Tools

The aim of this section is to clarify what is required and how to submit, grade and assess each assessment tool:

#### Exams:

This assessment tool is composed of one final exams, (30%), and a Midterm exam (20%). The final exam covers

all the learning outcomes of the course and covers the materials of the all chapters.

**Class activity:**

The **Class activity** assessment component is composed of three **Class activities** which cover all the CLOs. This assessment tool aims at enabling the students to practice data science topics covered within the course materials.

**Project:**

The term project is an indispensable tool for assessing student's ability to work in a team project throughout the semester, covering all the practical topics discussed in the weekly schedule. It measures the student's ability to collaborate and be an active learner through an enhanced project-based learning experience.

**Teaching and Learning Resources:**

**Textbook(s):**

Adhikari, A., DeNero, J. and Wagner, D.. (2023). Computational and Inferential Thinking: The Foundations of Data Science. 2 ed. UC Berkeley, Data 8..

**Recommended Readings:**

Vijay Kotu, Bala Deshpande. (2018). Data Science: Concepts and Practice. 2 ed. Morgan Kaufmann.

**Other Resources:**

**Recommended Readings:**

Other books in the field of data science (additional books may be recommended by Instructor).

1. Vijay Kotu, Bala Deshpande. (2018). Data Science: Concepts and Practice. 2nd Edition. Morgan Kaufmann ISBN: 9780128147627
2. VanderPlas, Jake. *Python data science handbook: Essential tools for working with data.* " O'Reilly Media, Inc.", 2016.

**Journals / articles**

Recent research paper in the field of data science (will be recommended by Instructor).

**Attendance policy:**

Attendance is compulsory. A student missing 25% of the total allocated course hours will be forced to withdraw in accordance with the university regulations.

**Plagiarism/Cheating:**

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

**Notes:**

**In-class Presentation:**

As this is a graduate course, students shall not be passive learners. A student shall select one of the suggested project topics, find at least four related papers, and then choose one of them for presentation after consulting

with the instructor. Each student is expected to make one in-class presentation. The presenter is expected to critique the paper by highlighting its weaknesses as well as its strong points. Moreover, the presenter should be prepared to answer detailed technical questions about the paper.

### **Project (research paper on a topic selected by the instructor):**

The objective of the project is to gain experience in research, implementation, testing, analysis, algorithm comparison, and enhancement of some of the most widely known data management and data science techniques (selected by the instructor).

Students are expected to write a high-quality research paper in one of the topics selected by the instructor. The project is divided into three parts. The combination of the three parts should result in a high-quality research paper of five to six pages at least.

Part 1: a 2-3 pages survey paper on the selected project topic. (Grade: 5%)

- Abstract
- Introduction
- Survey
- Future directions
- Conclusion

Part 2: Implementation, testing, and analysis of one of the techniques in the survey paper. (Grade: 25%)

- Implement the algorithm using Python.
- Testing the algorithm using a test data.

Part 3: Do any of the following tasks. (Grade: 5%)

- Test your code on real data and compare with experimental results of the paper.
- Team up with a classmate and merge the two algorithms and make a hybrid-algorithm (combining both algorithms in such a way to enhance performance). Both students must have finished Part 2.
- Suggest an improvement to the algorithm, and then upon approval from instructor implement and test the improved code.

**Each group then need to merge all the work into one original research paper and submit it to the instructor for approval and evaluation. Students are expected to do original work and produce novel results, comparing their work to at least one similar work from the related state-of-art and applying their algorithms (or solutions) to at least two related data sets!**

### **Grading rubrics:**

| <b>Grading Rubric for Course Projects</b> |               |                                                                 |                                                                                                     |                                                                                  |                                                                                               |
|-------------------------------------------|---------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Scaled Score:</b>                      |               | <b>1/0</b>                                                      | <b>2</b>                                                                                            | <b>3</b>                                                                         | <b>4</b>                                                                                      |
| <b>Criteria</b>                           | <b>Weight</b> | <b>Below Standard</b>                                           | <b>Approaching Standard</b>                                                                         | <b>At Standard</b>                                                               | <b>Exceeds Standard</b>                                                                       |
| Introduction                              | 10%           | No clear introduction, absent, or no evidence.                  | The introduction states the main topic but does not adequately preview the structure of the report. | The introduction states the main topic and previews the structure of the report. | The introduction is engaging, states the main topic and previews the structure of the report. |
| Data                                      | 20%           | Data are inadequate and selected following the wrong procedure. | Data are selected but procedure of data selection is not followed.                                  | Data selected following the standard procedure but not described in report.      | Data are correctly selected following standard procedure and adequately described in report.  |
| Analysis and                              | 20%           | Calculated wrong                                                | Calculated correct                                                                                  | Calculated correct                                                               | Calculated correct                                                                            |

| Grading Rubric for Course Projects   |        |                                                                          |                                                                                              |                                                                  |                                                                          |
|--------------------------------------|--------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------|
| Scaled Score:                        |        | 1/0                                                                      | 2                                                                                            | 3                                                                | 4                                                                        |
| Criteria                             | Weight | Below Standard                                                           | Approaching Standard                                                                         | At Standard                                                      | Exceeds Standard                                                         |
| Reporting of Results (if applicable) |        | measures and analysis not relevant.                                      | measures but analyses are not correct.                                                       | measures but analyses are not adequate.                          | measures and analyses are correct and provide more insights.             |
| Authenticity and Citation*           | 20-30% | Some contents are paraphrased; plagiarism is most likely.                | Some content is paraphrased; plagiarism is likely.                                           | Some content is paraphrased and no suspicion of plagiarism.      | All contents are original and well written.                              |
| Writing                              | 20-30% | Errors in language that cause the content to be mostly incomprehensible. | Some errors in language that often inhibit comprehension.                                    | Some errors in language that do not often inhibit comprehension. | Almost no error in language and well written.                            |
| Teamwork                             | 10%    | Team did not collaborate or communicate well.                            | Team did not collaborate or communicate well. A lack of mutual respect was frequently noted. | Team did collaborate or communicate well.                        | Team did collaborate or communicate well with high level mutual respect. |

| Grading Rubric for Course Assignments |        |                             |                                                                                                         |                                                                                          |                                                                                                                       |
|---------------------------------------|--------|-----------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Scaled Score:                         |        | 0/1                         | 2                                                                                                       | 3                                                                                        | 4                                                                                                                     |
| Criteria                              | Weight | Below Standard              | Approaching Standard                                                                                    | At Standard                                                                              | Exceeds Standard                                                                                                      |
| Focus: Purpose                        | 10%    | No awareness                | Shows limited awareness of purpose.                                                                     | Shows awareness of purpose.                                                              | Purpose is clear.                                                                                                     |
| Main idea                             | 10%    | No main idea                | Vague sense of a main idea, weakly supported throughout the paper.                                      | There is a main idea supported throughout most of the paper.                             | Clearly presents a main idea and supports it throughout the paper.                                                    |
| Organization: Overall                 | 5%     | No sense of organization    | There is a sense of organization, although some of the organizational tools are used weakly or missing. | Good overall organization, includes the main organizational tools.                       | Well-planned and well-thought out. Includes title, introduction, statement of main idea, transitions, and conclusion. |
| Organization: Paragraphs              | 5%     | Paragraphs lack clear ideas | Some paragraphs have clear ideas, support from examples may be missing and transitions are weak.        | Most paragraphs have clear ideas, are supported with some examples and have transitions. | All paragraphs have clear ideas, are supported with examples and have smooth transitions.                             |
| Content                               | 50-60% | Content is not sound        | Content is sound and solid; ideas are present but not particularly developed or                         | Well-presented and argued; ideas are detailed, developed and supported with              | Exceptionally well-presented and argued; ideas are detailed, well-                                                    |

| Grading Rubric for Course Assignments                  |     |                                                                                                                                                                                                 |                                                                                                                                                                                               |                                                                                                                                                         |                                                                                                                                                                |
|--------------------------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scaled Score:                                          |     | 0/1                                                                                                                                                                                             | 2                                                                                                                                                                                             | 3                                                                                                                                                       | 4                                                                                                                                                              |
|                                                        |     |                                                                                                                                                                                                 | supported; some evidence, but usually of a generalized nature.                                                                                                                                | evidence and details, mostly specific.                                                                                                                  | developed, supported with specific evidence and facts, as well as examples and specific details.                                                               |
| Research (if assignment includes a research component) | 10% | The paper does not use adequate research or if it does, the sources are not integrated well. They are not cited correctly according to MLA style, nor listed correctly on the Works Cited page. | Sources support some claims made in the paper, but might not be integrated well within the paper's argument. There may be a few errors in MLA style.                                          | Sources are well integrated and support the paper's claims. There may be occasional errors, but the sources and Works Cited conform to MLA style sheet. | Sources are exceptionally well-integrated and they support claims argued in the paper very effectively. Quotations and Works Cited conform to MLA style sheet. |
| Style: Sentence structure, word, choice, tone          | 5%  | Sentences aren't clear                                                                                                                                                                          | Sentences are generally clear word usage generally appropriate, details are somewhat adequate but may have awkward structure or unclear content; there may be patterns of punctuation errors. | Sentences are clear, word usage appropriate, details are enough but may lack variation; a few may be awkward and there may be a few punctuation errors. | Sentences are clear, word usage appropriate; details are enough and varied in pattern, from simple to complex, with excellent use of punctuation.              |
| Grammar and Mechanics                                  | 5%  | Continuous errors                                                                                                                                                                               | Shows a pattern of errors in spelling, grammar, syntax and/or punctuation. Could also be a sign of lack of proof-reading.                                                                     | A few errors in grammar, spelling, syntax and punctuation, but not many.                                                                                | Excellent grammar, spelling, syntax and punctuation.                                                                                                           |

Adapted from: <http://www.aur.edu/wp-content/uploads/2010/12/University-Writing-Rubric>

#### Grading Rubrics for Assessed Elements:

| Grading Rubric for Course Assignments |                         |                                    |                                    |                              |                                    |                       |                |
|---------------------------------------|-------------------------|------------------------------------|------------------------------------|------------------------------|------------------------------------|-----------------------|----------------|
| Scaled Score:                         |                         | F(0)                               | D(1)                               | C(2)                         | C+(3)                              | B(4)                  | A(5)           |
| Weight                                | Unacceptable            | V. Poor                            | Poor                               | Average                      | Good                               | Excellent             | Weighted Score |
| Focus: Purpose 10%                    | No awareness of purpose | Shows general awareness of purpose | Shows limited awareness of purpose | Purpose is clear but general | Purpose is clear but needs n focus | Purpose is very clear |                |

| Grading Rubric for Course Assignments                         |                                                                                                                                                                                            |                                                                                                                                                                             |                                                                                                                                           |                                                                                                                                                     |                                                                                                                                                   |                                                                                                                                                                  |                |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Scaled Score:                                                 | F(0)                                                                                                                                                                                       | D(1)                                                                                                                                                                        | C(2)                                                                                                                                      | C+(3)                                                                                                                                               | B(4)                                                                                                                                              | A(5)                                                                                                                                                             |                |
| Weight                                                        | Unacceptable                                                                                                                                                                               | V. Poor                                                                                                                                                                     | Poor                                                                                                                                      | Average                                                                                                                                             | Good                                                                                                                                              | Excellent                                                                                                                                                        | Weighted Score |
| <b>Main idea</b>                                              | 10%<br>No main idea                                                                                                                                                                        | Vague sense of main idea, weak support throughout the paper                                                                                                                 | There is a main idea supported throughout some parts of the paper                                                                         | Main idea is supported throughout the paper but needs focus                                                                                         | Main idea is clear, Good, supports it throughout the paper                                                                                        | Clearly presents a main idea and supports it throughout the paper                                                                                                |                |
| <b>Organization: Overall</b>                                  | 5%<br>No sense of organization                                                                                                                                                             | Lacking of good organizations in most parts                                                                                                                                 | Needs improvement throughout the paper                                                                                                    | There is a sense of organization, although some organizational tools are used weakly or misused                                                     | Good overall organization includes the main organizational tools                                                                                  | Well-planned and well-thought out. Includes title, introduction, statement of main idea, transitions, and conclusion                                             |                |
| <b>Organization: Paragraphs</b>                               | 5%<br>Paragraphs lack clear ideas                                                                                                                                                          | All paragraphs have limited support                                                                                                                                         | Some paragraphs have clear ideas but support from examples may be missing and transitions are weak                                        | Most paragraphs have clear ideas and are supported but need more examples and transitions                                                           | All paragraphs have clear ideas and are supported with examples and transitions                                                                   | All paragraphs have clear ideas, are supported with examples and have smooth transitions                                                                         |                |
| <b>Content</b>                                                | 50%<br>Content is not sound                                                                                                                                                                | Content is vague, unclear. Ideas are undeveloped or unsupported; no evidence to support it                                                                                  | Content is somehow solid but ideas need to be developed more clearly                                                                      | Content is sound and solid; ideas are present but not particularly developed or supported; some evidence, but usually of a generalized nature.      | Well-presented and argued, ideas are detailed, developed, supported with evidence and details, most specific.                                     | Exceptionally well-presented and argued; ideas are detailed, well-developed, supported with specific evidence & facts, as well as examples and specific details. |                |
| <b>Research (if assignment includes a research component)</b> | 10%<br>The paper does not use adequate research or if it does, the sources are not integrated well. They are not cited correctly according to MLA style, not listed correctly on the Works | Sources support limited claims made in the paper but they are not integrated well within the paper's argument. There are many errors in MLA style, not conform to the Works | Sources are not well integrated and support the paper's claims. There may be occasional errors but the sources conform to MLA style sheet | Sources support some claims made in the paper, but might not be integrated well within the paper's argument. There may be a few errors in MLA style | Sources are integrated and support the paper's claims. There may be occasional errors, but the sources and Works Cited conform to MLA style sheet | Sources are exceptionally well-integrated and support the paper very effectively. Quotations and Works Cited conform to MLA style sheet                          |                |

| Grading Rubric for Course Assignments                |        |                                     |                                                                                                                          |                                                                                                                       |                                                                                                                                                                                  |                                                                                                                                   |                                                                                                                                                  |                |
|------------------------------------------------------|--------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Scaled Score:                                        |        | F(0)                                | D(1)                                                                                                                     | C(2)                                                                                                                  | C+(3)                                                                                                                                                                            | B(4)                                                                                                                              | A(5)                                                                                                                                             |                |
|                                                      | Weight | Unacceptable                        | V. Poor                                                                                                                  | Poor                                                                                                                  | Average                                                                                                                                                                          | Good                                                                                                                              | Excellent                                                                                                                                        | Weighted Score |
|                                                      |        | Cited page                          |                                                                                                                          |                                                                                                                       |                                                                                                                                                                                  |                                                                                                                                   |                                                                                                                                                  |                |
| <b>Style: Sentence structure, word, choice, tone</b> | 5%     | Sentences are clear                 | Sentences are not clear, word usage is inappropriate, details are somewhat inadequate                                    | Some sentences are not clear, word usage is appropriate, details are not enough and there are many punctuation errors | Sentences are generally clear word usage generally appropriate, details are somewhat adequate but have awkward structure or unclear content there may be some punctuation errors | Sentences are clear, word usage appropriate; details are enough but lack variation and may be awkward with few punctuation errors | Sentences are clear, word usage appropriate; details are enough and varied in pattern, from simple to complex, with excellent use of punctuation |                |
| <b>Grammar &amp; Mechanics</b>                       | 5%     | Full of grammatical spelling errors | Shows a pattern of errors in spelling, grammar, syntax and/or punctuation. Could also be a sign of lack of proof-reading | Many errors in spelling, grammar, syntax and/or punctuation.                                                          | Some errors in spelling, grammar, syntax and/or punctuation                                                                                                                      | A few errors in grammar, spelling, syntax and punctuation                                                                         | Excellent grammar, spelling, syntax and punctuation                                                                                              |                |
| <b>Total Score</b>                                   | 100%   |                                     |                                                                                                                          |                                                                                                                       |                                                                                                                                                                                  |                                                                                                                                   |                                                                                                                                                  |                |
| <b>Scaled Score (divided by *)</b>                   |        |                                     |                                                                                                                          |                                                                                                                       |                                                                                                                                                                                  |                                                                                                                                   |                                                                                                                                                  |                |

Adapted from (weights modified):

<http://www.aur.edu/wp-content/uploads/2010/12/University-Writing-Rubric-Approved-by-Curriculum-Committee-Feb-2010.doc>

| Grading Rubric for Presentations  |         |                                         |                                                |                                        |                                          |                                 |                                 |                |
|-----------------------------------|---------|-----------------------------------------|------------------------------------------------|----------------------------------------|------------------------------------------|---------------------------------|---------------------------------|----------------|
| Scaled Score:                     |         | F(0)                                    | D(1)                                           | C(2)                                   | C+(3)                                    | B(4)                            | A(5)                            |                |
|                                   | Weights | Unacceptable                            | V. Poor                                        | Poor                                   | Average                                  | Good                            | Excellent                       | Weighted Score |
| <b>Coherence and Organization</b> | 15%     | Audience cannot understand presentation | Audience has difficulty following presentation | Student presents information in a poor | Student presents information in logical, | Student presents information in | Student presents information in |                |

| Grading Rubric for Presentations      |            |                                                                                                          |                                                                                                 |                                                                                                                                   |                                                                                                                                                   |                                                                                                                   |                                                                                                                               |                |
|---------------------------------------|------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------|
| Scaled Score:                         |            | F(0)                                                                                                     | D(1)                                                                                            | C(2)                                                                                                                              | C+(3)                                                                                                                                             | B(4)                                                                                                              | A(5)                                                                                                                          |                |
|                                       | Weights    | Unacceptable                                                                                             | V. Poor                                                                                         | Poor                                                                                                                              | Average                                                                                                                                           | Good                                                                                                              | Excellent                                                                                                                     | Weighted Score |
|                                       |            | because there is no logical flow of ideas.                                                               | because student jumps around.                                                                   | logical sequence which audience cannot follow                                                                                     | interesting sequence which audience can follow.                                                                                                   | logical sequence which audience can follow.                                                                       | logical, interesting sequence which audience can follow.                                                                      |                |
| <b>Subject Knowledge</b>              | <b>40%</b> | Student does not have grasp of information; student cannot answer questions about subject                | Student is uncomfortable with information and is able to answer only rudimentary questions      | Student answers the questions poorly and fails to elaborate                                                                       | Student demonstrates adequate knowledge by answering some class questions with explanations and elaboration.                                      | Student is at ease with expected answers to all questions, but fails to elaborate.                                | Student demonstrates full knowledge (more than required) by answering all class questions with explanations and elaboration.  |                |
| <b>Quality of Content and Visuals</b> | <b>15%</b> | Student's presentation has many spelling errors and/or grammatical errors. Student does not use graphics | Presentation has some misspellings and/or grammatical errors. Student uses superfluous graphics | Presentation has many misspellings and/or grammatical errors. Some of the Student's graphics are relate to text and presentation. | Presentation has occasional misspellings or grammatical errors. Student's graphics explain and reinforce most of the screen text and presentation | Presentation has a few misspellings and/or grammatical errors. Student's graphics relate to text and presentation | Presentation has no misspellings or grammatical errors. Student's graphics explain and reinforce screen text and presentation |                |
| <b>Posture and Eye Contact</b>        | <b>15%</b> | Student reads all of report with no eye contact                                                          | Student rarely uses eye contact, but still reads most                                           | Student maintains eye contact occasionally but                                                                                    | Student maintains eye contact with audience                                                                                                       | Student maintains eye and returns to notes                                                                        | Student maintains eye contact with                                                                                            |                |

| Grading Rubric for Presentations   |            |                                                                                                                 |                                                                                                                     |                                                                                                                                |                                                                                                                  |                                                                                                                |                                                                                                                            |                |
|------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------|
| Scaled Score:                      |            | F(0)                                                                                                            | D(1)                                                                                                                | C(2)                                                                                                                           | C+(3)                                                                                                            | B(4)                                                                                                           | A(5)                                                                                                                       |                |
|                                    | Weights    | Unacceptable                                                                                                    | V. Poor                                                                                                             | Poor                                                                                                                           | Average                                                                                                          | Good                                                                                                           | Excellent                                                                                                                  | Weighted Score |
|                                    |            |                                                                                                                 | of report                                                                                                           | frequently returns to notes                                                                                                    | most of the time, returning to notes occasionally                                                                | rarely                                                                                                         | audience, without returning to notes                                                                                       |                |
| <b>Elocution</b>                   | <b>15%</b> | Student mumbles, incorrectly pronounces terms, and speaks too quietly for students in the back of class to hear | Student's voice is low. Student incorrectly pronounces terms. Audience members have difficulty hearing presentation | Student's voice is not clear. Student pronounces some of the words incorrectly. Most audience members cannot hear presentation | Student uses a clear but low voice, precise pronunciation of terms. Some audience members can hear presentation. | Student's voice is clear. Student pronounces most words correctly. Most audience members can hear presentation | Student uses a clear voice and correct, precise pronunciation of terms so that all audience members can hear presentation. |                |
| <b>Total Score</b>                 | 100%       |                                                                                                                 |                                                                                                                     |                                                                                                                                |                                                                                                                  |                                                                                                                |                                                                                                                            |                |
| <b>Scaled Score (divided by *)</b> |            |                                                                                                                 |                                                                                                                     |                                                                                                                                |                                                                                                                  |                                                                                                                |                                                                                                                            |                |

Adapted from (weights are adjusted):

<http://www.ncsu.edu/midlink/rub.pres.html>

| Grading Rubric for Course Project (Technical) |        |                                                 |                                                                           |                                                                                 |                                                                         |                                                                            |                                                                          |                |
|-----------------------------------------------|--------|-------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------|----------------|
| Scaled Score:                                 |        | F(0)                                            | D(1)                                                                      | C(2)                                                                            | C+(3)                                                                   | B(4)                                                                       | A(5)                                                                     |                |
|                                               | Weight | Unacceptable                                    | V. Poor                                                                   | Poor                                                                            | Average                                                                 | Good                                                                       | Excellent                                                                | Weighted Score |
| <b>Contribution</b>                           | 25%    | No idea of what requirements and objectives are | No more than half of the requirements identified, evaluated and completed | All requirements identified and evaluated but some objectives are not completed | All requirements are identified and evaluated but some missing elements | No more than half of the requirements identified, evaluated and completed. | All requirements and objectives are identified, evaluated and completed. |                |
| <b>Team Work</b>                              | 25%    | No team work                                    | Team did not work in a team                                               | The team worked but not together most of the time                               | The team worked well together                                           | The team worked well together                                              | The team worked well together to achieve                                 |                |

| Grading Rubric for Course Project (Technical) |                                        |                                                                             |                                                                                                                                                         |                                                                                                                                                                                                                          |                                                                                                                                                   |                                                                                                                                                                                                                                          |                |
|-----------------------------------------------|----------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Scaled Score:                                 | F(0)                                   | D(1)                                                                        | C(2)                                                                                                                                                    | C+(3)                                                                                                                                                                                                                    | B(4)                                                                                                                                              | A(5)                                                                                                                                                                                                                                     |                |
| Weight                                        | Unacceptable                           | V. Poor                                                                     | Poor                                                                                                                                                    | Average                                                                                                                                                                                                                  | Good                                                                                                                                              | Excellent                                                                                                                                                                                                                                | Weighted Score |
|                                               | demonstrated                           |                                                                             | communicate well. Some members would work independently, without regard to objectives or priorities. A lack of respect and regard was frequently noted. | the time, with only a few occurrences of communication breakdown or failure to collaborate when appropriate. Members were mostly respectful of each other                                                                | objectives. Each member contributed in a valuable way to the project. All data sources indicated a high level of mutual respect and collaboration |                                                                                                                                                                                                                                          |                |
| <b>Subject knowledge</b>                      | 20%<br>No understanding of the subject | Serious issues with the understanding                                       | The deliverable did not demonstrate knowledge of the course content, evidence of the research effort, or depth of thinking about the topic              | The deliverable demonstrated knowledge of the course content by integrating major concepts into the response. The deliverable also demonstrated evidence of limited research effort and/or initial of thinking about the | Not more than one serious issue                                                                                                                   | The deliverable demonstrated knowledge of the course content by integrating major and minor concepts into the response. The deliverable also demonstrated evidence of extensive research effort and a depth of thinking about the topic. |                |
| <b>Supporting Material</b>                    | 20%<br>No supporting material          | Very little supporting material                                             | Insufficient information was obtained and/or sources lack validity. Analysis and design considerations were not supported by the information collected  | Sufficient information was obtained and most sources were valid. Analysis and design considerations were mostly supported by the information.                                                                            | Not more than one issue with regards to supporting material                                                                                       | All relevant information was obtained and information sources were valid. Analysis and design considerations were well supported by the information                                                                                      |                |
| <b>Composition</b>                            | 10%<br>No structure                    | Almost impossible to understand the structure and composition of the report | The deliverable lacked overall organization. The reader had to make considerable effort to understand the flow of ideas. Diagrams were                  | The deliverable was well organized and clearly written for the most part. In some areas logic and/or flow of ideas were difficult to follow. Words were well chosen with some minor                                      | Not more than 3 minor issues                                                                                                                      | The deliverable was well organized and clearly written. The underlying logic was clearly articulated and easy to follow. Words were chosen that precisely expressed the intended meaning and                                             |                |

| Grading Rubric for Course Project (Technical) |              |         |                                                                                                                                                                                                                                                                                                                    |         |       |                                                                                                                                                     |                |  |
|-----------------------------------------------|--------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--|
| Scaled Score:                                 |              | F(0)    | D(1)                                                                                                                                                                                                                                                                                                               | C(2)    | C+(3) | B(4)                                                                                                                                                | A(5)           |  |
| Weight                                        | Unacceptable | V. Poor | Poor                                                                                                                                                                                                                                                                                                               | Average | Good  | Excellent                                                                                                                                           | Weighted Score |  |
|                                               |              |         | absent or inconsistent with the text. Grammatical and spelling errors made it difficult for the reader to interpret the text in place of expectations. Diagrams were inconsistent with the text. Sentences were mostly grammatical and only a few spelling errors were present but they did not hinder the reader. |         |       | supported reader comprehension. Diagrams or analyses enhanced and clarified presentation of ideas. Sentences were grammatical and free from errors. |                |  |
| Total Score                                   | 100%         |         |                                                                                                                                                                                                                                                                                                                    |         |       |                                                                                                                                                     |                |  |
| Scaled Score (divided by *)                   |              |         |                                                                                                                                                                                                                                                                                                                    |         |       |                                                                                                                                                     |                |  |

Comments:

---



---



---